

Nick Shum

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EDUCATION

University of Waterloo | Bachelor's of Computer Engineering, Co-op
Waterloo, ON

Apr. 2027

- Relevant Coursework: Data Structures & Algorithms, Systems Programming & Concurrency, Operating Systems, Networking, Database Systems, Compilers

SKILLS

Languages: TypeScript, JavaScript, Java, Python, C/C++, SQL, HTML, CSS

Frameworks/Libraries: React, Next.js, Node.js, Flask, Django, Spring Boot, Tailwind CSS

Cloud/DevOps: Docker, Kubernetes, SQL Server, PostgreSQL, MongoDB, AWS, Azure, Firebase, Vercel

WORK EXPERIENCE

CharacterQuilt | Software Engineer (Contract)
Waterloo, ON

May 2025 - Aug. 2025

- Eliminated API timeouts and slashed server response time by **99.3%** (30 to 0.2s) by offloading heavy processing to an asynchronous job queue using **Python, AWS Lambda**, and **SQS**
- Deployed a persona-driven content editing pipeline, reducing human review time by **72.2%** (18 to 5 min) using Pinecone and GPT-4o prompt chaining
- Shipped an AI agent with context retrieval and validation to generate accurate table columns/filters, cutting **~14 hours** of manual query work weekly using Python and OpenAI API

CharacterQuilt | Software Engineer Intern
Denver, CO

Jan. 2025 - Apr. 2025

- Reduced initial page render by **86.6%** (22 to 3s) by lazy-loading large React components and batch executing asynchronous fetch requests
- Developed a serverless event notification platform using AWS EventBridge, DynamoDB, and Perplexity API to deliver weekly alerts on target company news and job posting indicators
- Overhauled the core product UI using **React** and **TypeScript** in a **fast-paced start-up**, giving early users a polished experience ahead of public launch

IBM | Software Engineer Intern
Toronto, ON

May 2024 - Aug. 2024

- Enabled natural language CSV querying by implementing retrieval-augmented generation (RAG) with LangChain, FAISS vector indexing, pandas, and Llama-3-70B LLM
- Delivered a data-processing package using Apache Spark to replace legacy processes, slashing runtime from **2 hours to 20 minutes**
- Engineered a rules-based anomaly detection engine to verify **\$10B+** in insurance claims using Python and SQL

Versaterm | Full Stack Engineer Intern
Vancouver, BC

Sep. 2023 - Dec. 2023

- Built and launched an employee organizational chart using **Spring Boot** and **Vue.js**, streamlining inter-departmental communication across **1000+** police agencies
- Led cross-team initiative to revamp and scale a geospatial incident tracking system with geofencing functionality using Vue.js and Leaflet

PROJECTS

🔗 **ResuMatch - Job Application Manager**, Firebase, AWS S3, React, Django, SQLite

- Developed a job search web application, enabling users to create accounts, track job applications, and link uploaded resumes to specific positions for optimized job search management
- Enabled encryption and access controls by leveraging **AWS S3** with **SQLite** for enhanced data security and retrieval in a relational database